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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE) - MG3DSECHEA01 - FUNDAMENTALS OF FORENSIC SCIENCE
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Instructions to Examiners:

- This scheme is only a guidance for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all their points in the answers of the students.

Part A

Short Answer Type Questions

Answer any 5 questions. (Each question marks carries 4)

1. Which act in India governs admissibility of expert opinion? (Correct answer: 4 marks)

In India, the admissibility of expert opinion is governed by the Indian Evidence Act, 1872 — in particular Section 45 which deals with the opinion of persons especially skilled in foreign law, science, art, handwriting or finger impressions.

2. What is the lowest level criminal court? (Correct answer: 4 marks)

Judicial Magistrate of the Second Class (JM II)

In Metropolitan areas, the equivalent lowest court is the Metropolitan Magistrate.

3. In how many hours must an arrested person be presented before a magistrate? (Correct answer: 4 marks)

Article 22(2) of the Constitution of India

Section 57, Code of Criminal Procedure (CrPC), 1973

The arrested person must be produced before the nearest magistrate within 24 hours, excluding the time necessary for the journey.

4. What are various psychotic disorders? (Correct answer: 4 marks)

Schizophrenia, Depression, Bipolar disorder, Delusional disorder, Psychotic disorder due to medical condition like traumatic brain injury or dementia, Substance induced psychotic disorder such as drugs or alcohol.

5. What is a white-collar crime? (Correct answer: 4 marks)

White-collar crime refers to nonviolent, financially motivated offenses that are typically committed by individuals, businesses, or government officials in positions of trust or authority. The term was first coined by sociologist Edwin Sutherland in 1939 to describe crimes committed by a person of respectability and high social status in the course of their occupation. Common examples include: fraud (securities, mortgage, healthcare, credit card, etc.), embezzlement, insider trading, bribery and corruption, money laundering, tax evasion, identity theft, cybercrime, and Ponzi or pyramid schemes.

6. What are cognitive assessments in forensic psychology? (Correct answer: 4 marks)

Cognitive assessment in forensic psychology evaluates a person's mental abilities like memory, attention and problem-solving, using standardized tests to help inform legal decisions. This can determine an individual's competency to stand trial, assess the impact of brain injury on behavior and evaluate the credibility of witness testimony. Assessments may also explore how cognitive processes contribute to offending behavior.

7. Which branch deals with poisons and drugs in the body? (Correct answer: 4 marks)

The branch of forensic science that deals with poisons and drugs in the body is known as forensic toxicology. This fi eld focuses on the detection and analysis of toxic substances, drugs, and poisons in biological samples, such as bloo

d, urine, and tissues. Forensic toxicologists play a crucial role in criminal investigations, workplace drug testing, and medical examinations, helping to establish the cause of death, determine drug involvement in criminal cases, and assess the impact of substances on behavior.

Part B

Short Essay Type Questions

Answer any 5 questions. (Each question carries 7 marks)

8. What is the role of forensic science in the Indian criminal justice system?

by providing scientific and objective support to crime investigation and evidence evaluation. It bridges the gap between science and law, helping courts reach accurate and fair judgments. The major roles of forensic science include: Scientific Crime Investigation, Collection and Preservation of Evidence, Scientific Examination of Evidence, Establishing Identity, Expert Testimony in Court, Crime Prevention and Criminal Profiling, Ensuring Fair Justice.

(Proper explanation with any four valid points: 5 marks)

9. Mention any two divisions of the FBI and their roles. (Any two divisions: 5 marks)

Criminal Investigative Division (CID), Laboratory Division (FBI Crime Lab)

10. List any two Fundamental Rights. (Any two Rights: 5 marks)

Right	Articles	Description
Right to Equality	14–18	Equality before law, prohibition of discrimination, abolition of untouchability & titles
Right to Freedom	19–22	Freedom of speech & expression, movement, assembly, profession; protection in respect of conviction & arrest
Right against Exploitation	23–24	Prohibits human trafficking, forced labor, and child labor in hazardous industries
Right to Freedom of Religion	25–28	Freedom of conscience and religion, manage religious institutions
Cultural and Educational Rights	29–30	Protects interests of minorities in preserving culture and establishing educational institutions
Right to Constitutional Remedies	32	Right to approach the Supreme Court for enforcement of Fundamental Rights (Habeas Corpus, Mandamus, etc.)

11. What types of compensation are available for victims of crime?

Court-ordered compensation under Section 357 of the Criminal Procedure Code (CrPC) requires offenders to pay fines or damages to victims. State Victim Compensation Schemes under Section 357A provide financial aid to victims and their dependents. The Central Victim Compensation Fund (CVCF) assists states in providing compensation for serious crimes such as rape, acid attacks, and human trafficking. Some states also provide interim relief for immediate medical or financial needs. Furthermore, the Supreme Court and High Courts can award compensation for violations of fundamental rights under Articles 32 and 226 of the Constitution.**(Any four valid points: 5 marks)**

12. Differentiate between modus operandi & signature in concept with forensic behavioral analysis.

The *modus operandi (MO)* refers to the method of operation, the techniques and procedures a criminal consistently uses to commit crimes. In contrast, a *signature* involves behaviors or details of the crime that are not necessary to accomplish it but fulfill the offender's psychological or emotional needs (for instance, specific ritualistic actions, posing of the body, or leaving a particular mark or object at the scene). Unlike the modus operandi, the signature tends to remain more fixed because it stems from the offender's inner fantasies or personality, and thus it can be a powerful clue in linking crimes to the same individual. **(Correct answer: 5 marks)**

13. Name any three characteristics of forensic science.

Forensic Science is the scientific discipline engaged in the recognition, identification, individualization, and evaluation of physical evidence by using the laws and principles of natural science to find answers to doubtful questions that may arise in a court of law. It encompasses various scientific disciplines and integrates them to support investigations, legal proceedings, and the administration of justice. The range of a forensics includes either one or all the three following major activities in analyzing and interpreting the physical evidences in the forensic science laboratories: Identification, Individualization & Reconstruction

(Proper explanation with any three valid points: 5 marks)

14. What does the DNA Fingerprinting Unit in State FSLs specialize in?

State FSLs in India specialize in using DNA fingerprinting to generate DNA profiles from various biological exhibits and compare them to link suspects to crime scenes, identify victims and resolve disputed relationships. State FSLs use DNA Analysis for a range of critical cases such as:

Violent crimes like murder, sexual assault, rape, Paternity and Maternity disputes, Identification of unidentified bodies, Mass disasters, Child trafficking, Identification from trace evidence, Wildlife crimes

(Proper explanation with any three valid points: 5 marks)

Part C

Essay Type Questions

Answer any 1 questions. (Each question carries 15 marks)

15. Discuss the educational pathways available for forensic scientists in India. Include entry-level qualifications, postgraduate specializations, and doctoral research opportunities.

Educational Pathways for Forensic Scientists in India

1) **Entry-Level (After Class 12):** Eligibility: 10+2 with Science (PCM/PCB), English preferred.

2) **Postgraduate Specializations (PG):** General eligibility: Relevant bachelor's degree (usually 50–60%+); subject prerequisites vary by program.

A) M.Sc. Forensic Science (2 years) Year 1 (core

B) M.Sc. subject-focused (2 years) M.Sc. Forensic Chemistry / Analytical Chemistry (Forensic focus), M.Sc. Toxicology / Pharmaceutical Chemistry (for poison/drug analysis), M.Sc. Cyber/Digital Forensics & Information Security

C) M.Tech / ME (2 years) (for tech candidates) Cyber Forensics & Information Security / Data Science for Security Eligibility: B.Tech/BE (CS/IT/EC) or M.Sc. CS/IT; usually GATE or institute exam.

D) PG Diplomas / Certificates (6–12 months) Good for working professionals or bridge-ups:

PGD in Forensic Document Examination, PGD in Cyber/Digital Forensics (incl. mobile forensics), PGD in DNA Forensics / Serology, PGD in Forensic Toxicology / Narcotics analysis

3) **Doctoral Research (Ph.D.):** Eligibility: Master's in Forensic Science/Chemistry/Biology/CS (as relevant).

(Reasonable explanation for entry-level qualifications (4 points) – 8 marks, postgraduate specializations (3 points) – 4 marks, and doctoral research opportunities – 3 marks)

16. Explain the major functions and duties of Government Examiner of Questioned Documents (GEQD).

Government Examiner of Questioned Documents (GEQD): Functions and Duties

The **Government Examiner of Questioned Documents (GEQD)** is a highly specialized forensic authority in India responsible for the scientific examination of documentary evidence. GEQD laboratories function under the Directorate of Forensic Science Services (DFSS), Ministry of Home Affairs, Government of India. Their findings support law-enforcement agencies and judicial authorities in resolving disputes related to written and printed materials.

Major Functions and Duties

1. Examination of Handwriting and Signatures
2. Analysis of Altered or Tampered Documents.
3. Examination of Ink, Paper, and Writing Instruments
4. Detection of Forgery and Counterfeiting
5. Typewriting, Printing, and Photocopy Examination
6. Seal, Stamp, and Impression Analysis
7. Expert Testimony in Courts
8. Advisory Role to Law-Enforcement Agencies
9. Research and Development
10. Maintenance of Case Records and Quality Standards

(Proper explanation with any eight valid points: 15 marks)